

Exercise 1: Probability Spaces and Distributions

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Problem 1 (Closedness of σ -algebra for countable intersections):

Consider a σ -algebra $\mathcal{F}$ and $A_1, A_2, A_3, \dots \in \mathcal{F}$. Prove that $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$.

Problem 2 (PDF and probability):

Consider a continuous random variable X with probability density function $f_X(x)$. Prove that $\int_a^b f_X(x) dx = \mathbb{P}(a \leq X \leq b)$.

Problem 3 (The Mode of a Binomial Distribution):

Let $X \sim \text{Binomial}(n, p)$. The value of $k \in \{0, 1, \dots, n\}$ that maximizes the probability mass function $\mathbb{P}(X = k)$ is called the **mode** of the distribution. Show that this value of k is an integer that satisfies $(n+1)p - 1 \leq k \leq (n+1)p$.

Hint: Consider the ratios $\frac{\mathbb{P}(X=k)}{\mathbb{P}(X=k-1)}$.

Problem 4 (PDF of the minimum of independent exponential random variables):

Consider two independent random variables, $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$. Let a new random variable Y be defined as the minimum of the two: $Y := \min(X_1, X_2)$.

Prove that Y also follows an exponential distribution, and find its rate parameter.

Hint: Find the CDF of Y , $F_Y(y) = \mathbb{P}(Y \leq y)$, by first calculating the “survival” function, $\mathbb{P}(Y > y)$.

Problem 5 (Probabilities in normal distribution):

Consider a random variable $X \sim N(\mu, \sigma^2)$. Calculate the probability $\mathbb{P}(\mu - \sigma \leq X \leq \mu + \sigma)$.

Hint: Use that $\int_{-\infty}^1 \frac{1}{\sqrt{2\pi}} e^{-t^2/2} \approx 0.8413$. This integral is computed via a numerical method.